Mount Kenya University

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BSCCS/2024/55349

BSC Computer Science

Unit Code: BIT3208 Advanced Web Design and Development

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Project Details

Project Title: Runi Empire Grooming Barbershop Management System

Selected Technology: JavaScript, PHP, MySQL, XAMPP

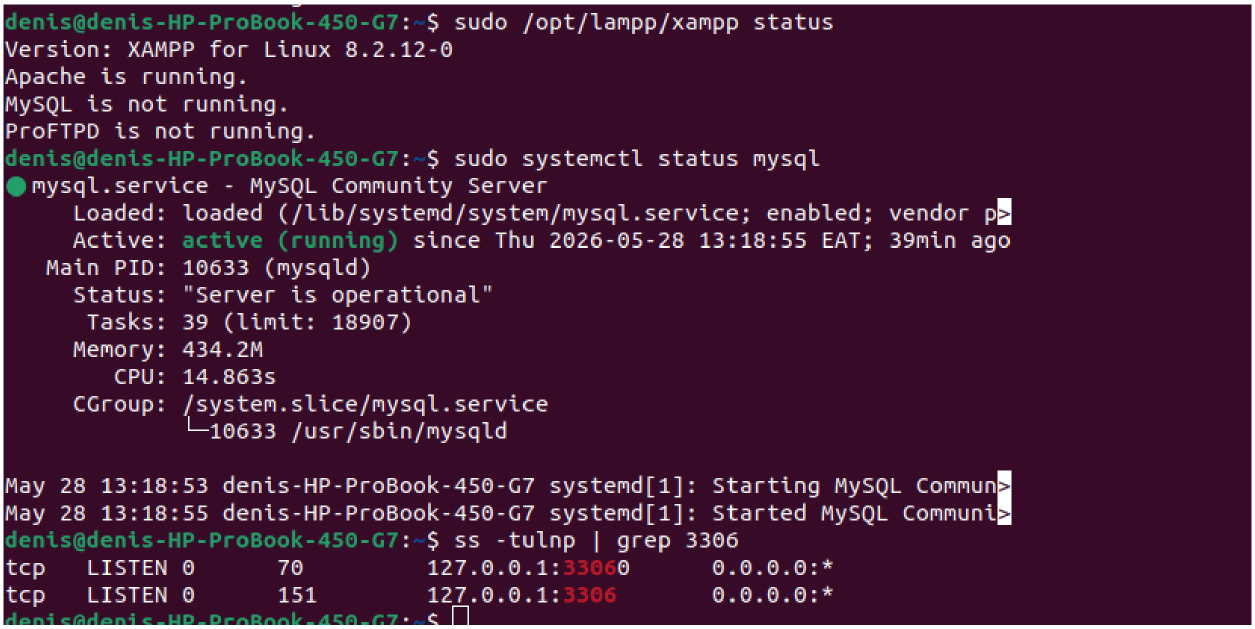
GitHub Link:

https://github.com/deniskibichiy/BIT3208_Advanced_Web_Design_and_Development#

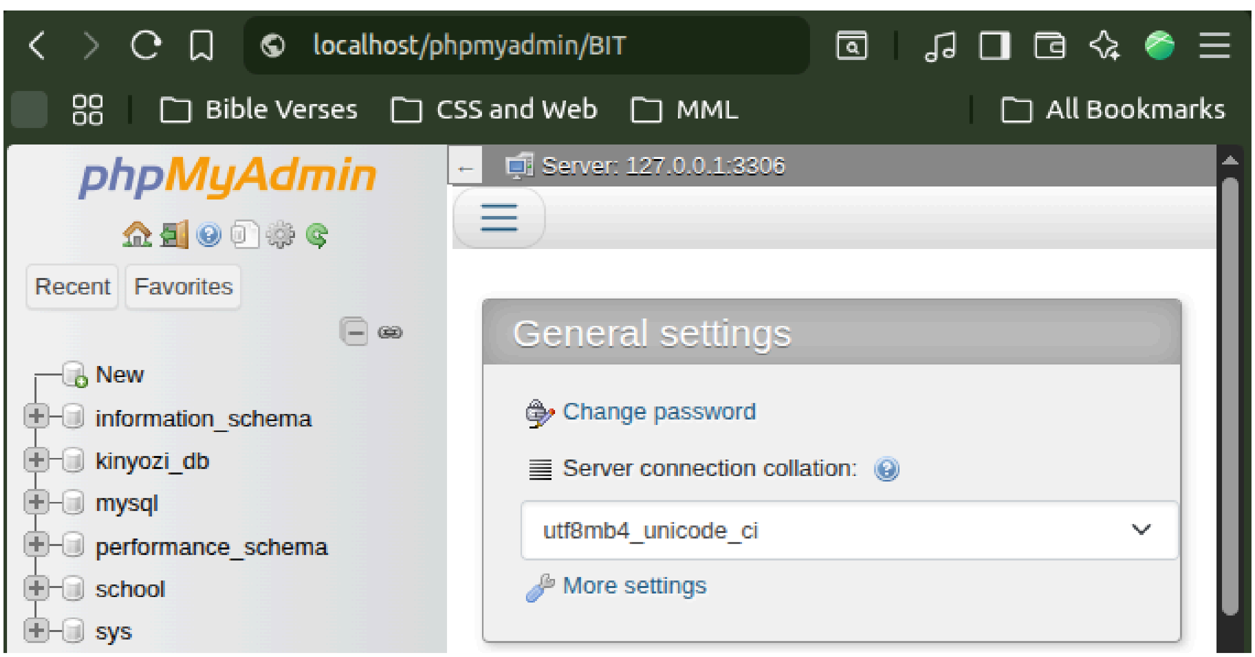
Table of Contents

1. Week 1: Environment setup and database connection

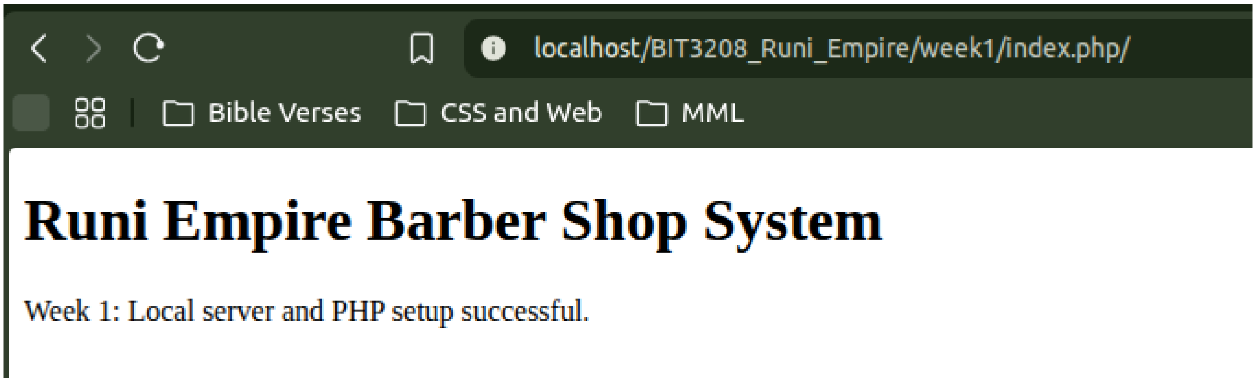
1.1. Figure 1: XAMPP Installation and Setup with MySQL in Linux



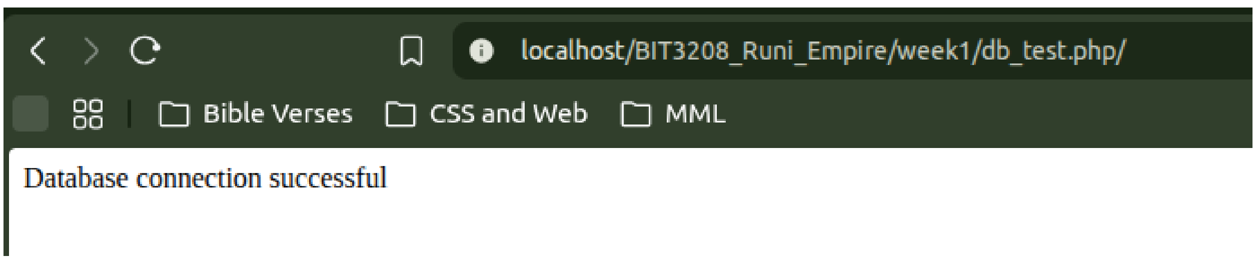
1.2. Figure 2: Apache + MySQL Running and Operational



1.3. Figure 3: Index.php successfully rendered



1.4. Figure 4: Database connection successful



1.5. Figure 5: Week1 database

```
mysql> USE week1db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+-------------------+
| Tables_in_week1db |
+-------------------+
| users             |
+-------------------+
1 row in set (0.01 sec)

mysql> DESCRIBE users
    -> ;
+--------------+--------------+------+-----+-------------------+----------
---------+
| Field        | Type         | Null | Key | Default           | Extra
         |
+--------------+--------------+------+-----+-------------------+----------
---------+
| id           | int          | NO   | PRI | NULL              | auto_incr
ement    |
| name         | varchar(100) | YES  |     | NULL              |
         |
| phone_number | varchar(20)  | YES  |     | NULL              |
         |
| created_at   | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_G
ENERATED |
+--------------+--------------+------+-----+-------------------+----------
---------+
4 rows in set (0.01 sec)
```

Week 1 reflection:

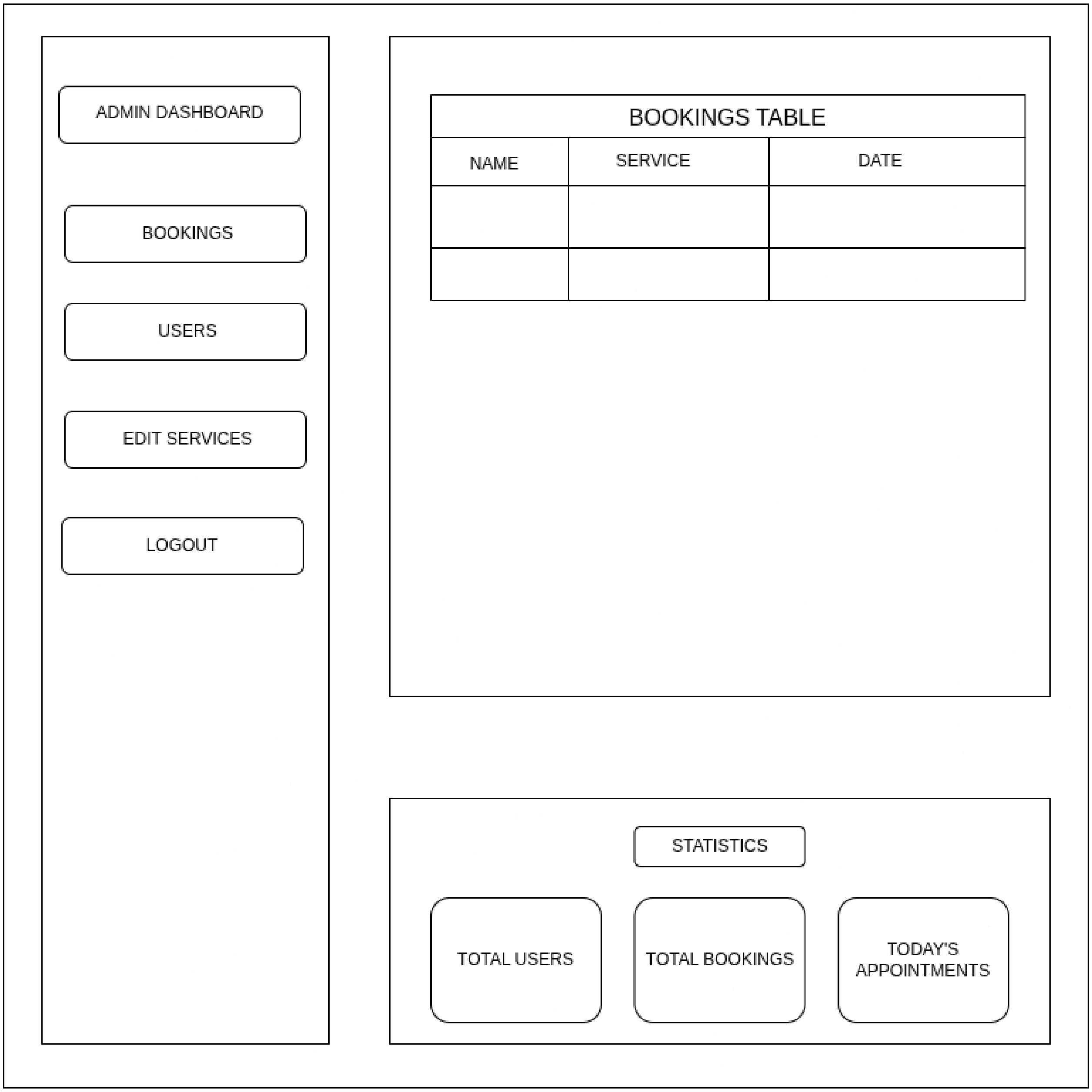
During week one, the focus was on setting up the environment to facilitate practical activities which are part of the Unit requirements. A key challenge I experienced is that the XAMPP MySQL couldn't load successfully due to port clashes with the system MySQL. To solve this, I had to configure Apache to use system MySQL instead of the one provided by PHPMyAdmin, and the process was successful.

2. Week 2: User Interface Planning and System Design

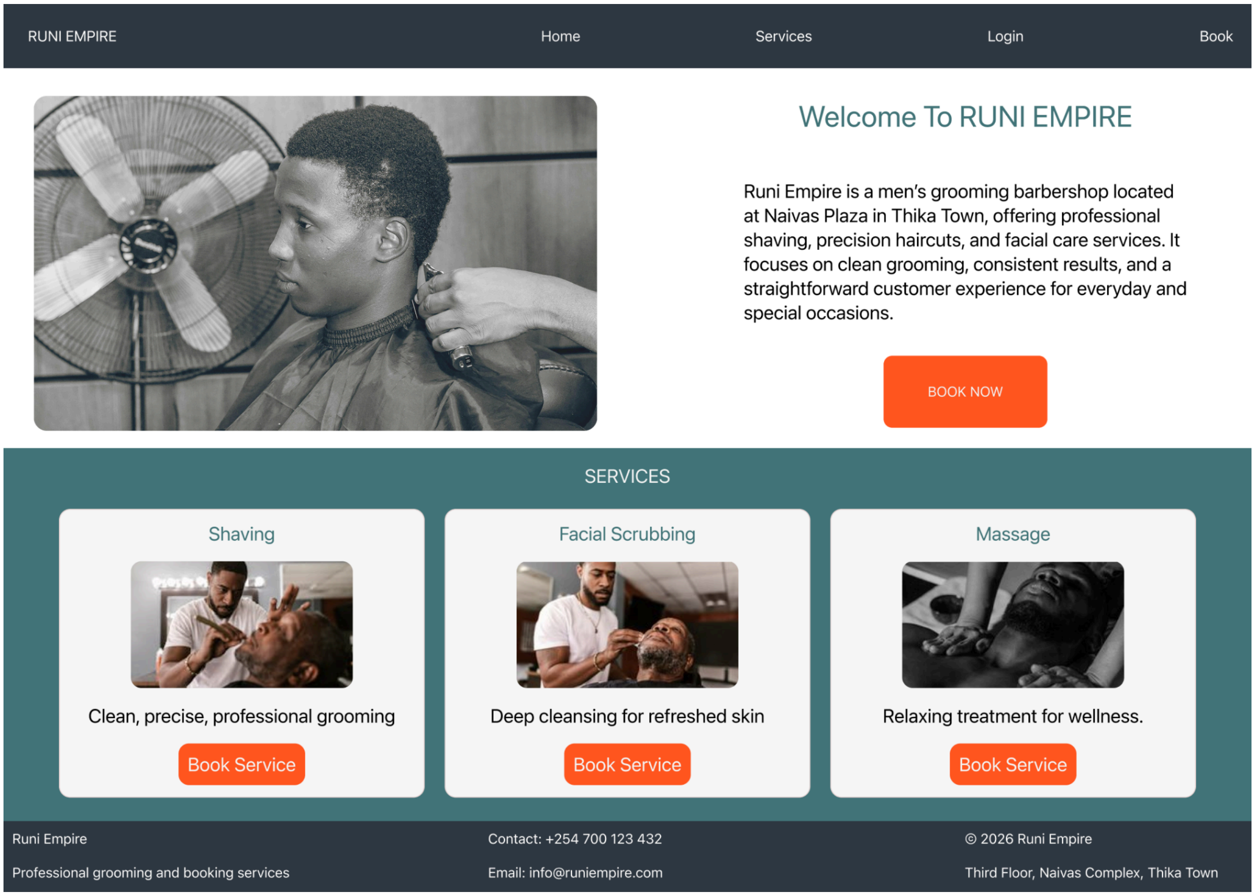
2.1. Figure 2.1: Homepage User Interface Mockup

2.2. Figure 2.2: Login Page User Interface Mockup

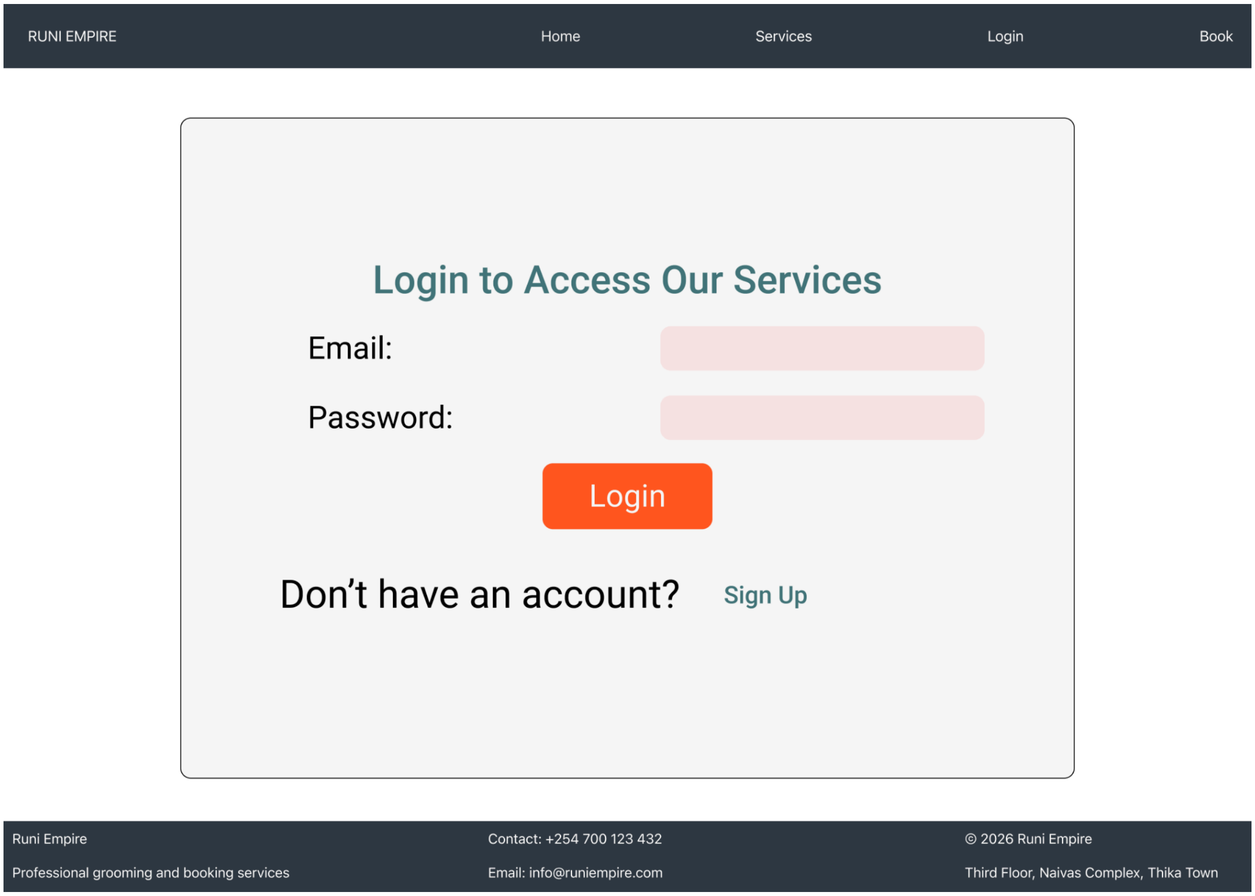
2.3. Figure 2.3: Admin Dashboard User Interface Mockup



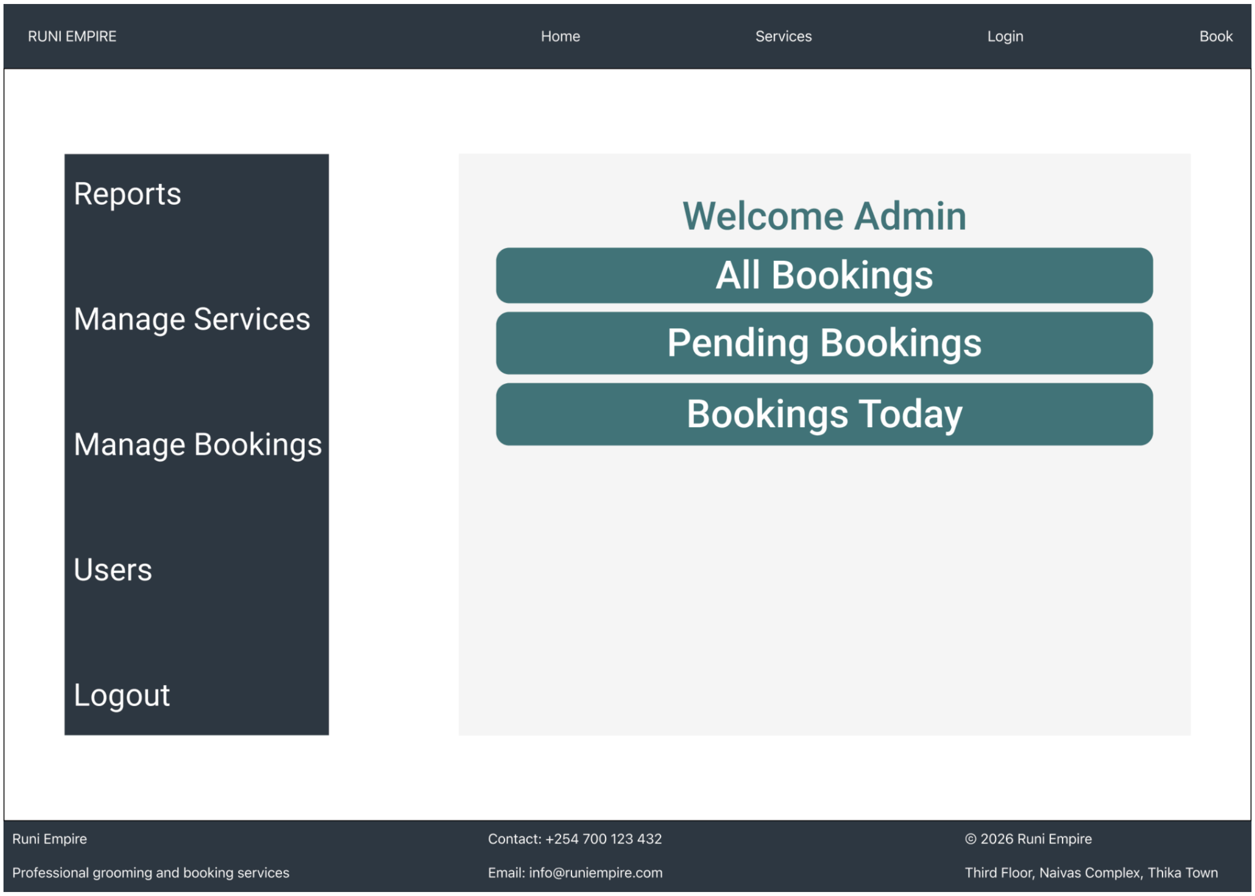
2.4. Figure 2.4: Homepage Figma prototype



2.5. Figure 2.5: Login page Figma prototype



2.6. Figure 2.6: Dashboard Figma prototype



2.7. Figure 2.7: Color Scheme

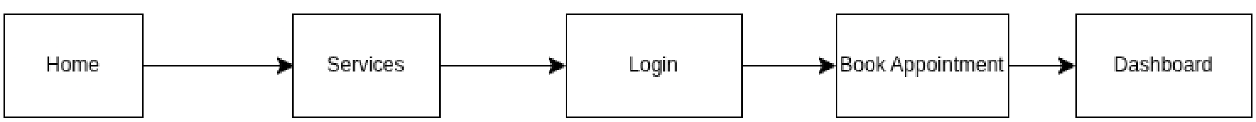
i. Background: #F5F5F5 : Light Gray

ii. Secondary/Highlights: #76ABAE: Teal

iii. Main text: #303841: Charcoal

iv. Accent/Actions: #FF5722: Orange

2.8. Figure 2.8: Navigation Structure

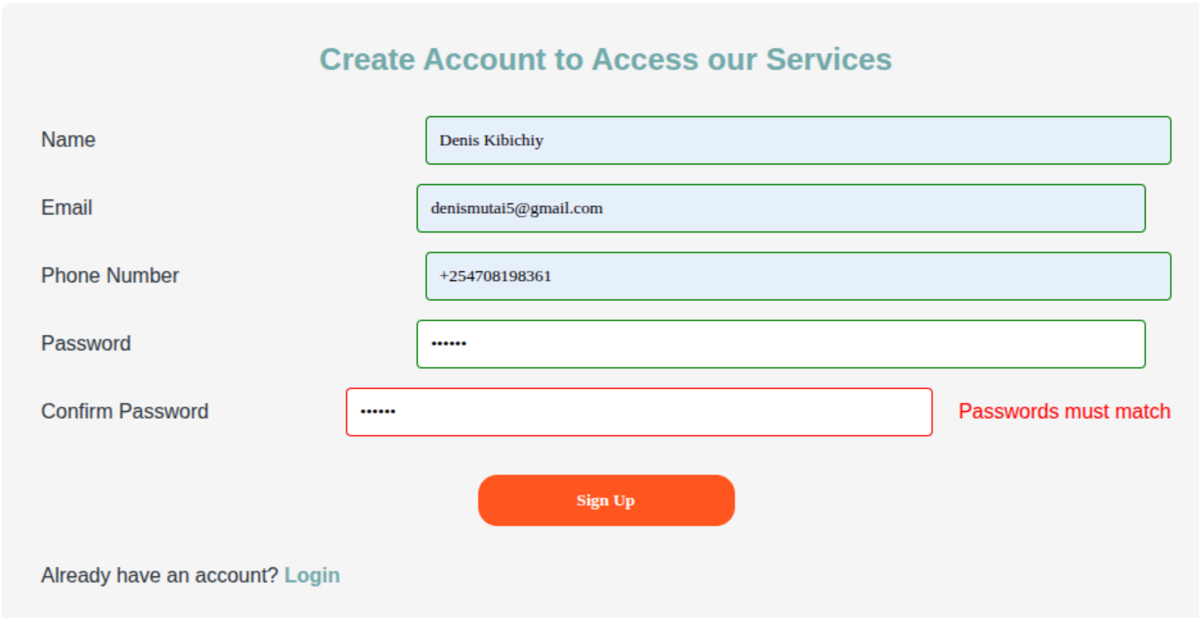


Reflection week 2:

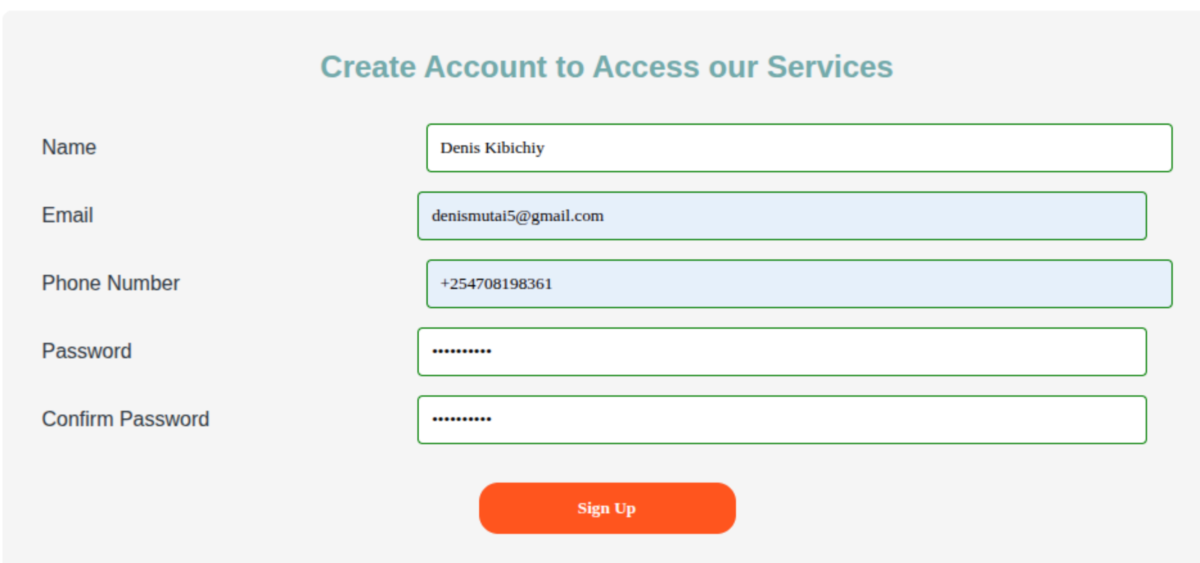
Week 2 provided me with another key challenge in that after visualizing how my website should be laid out, the next obstacle was designing the prototype using Figma. To complete the process successfully, I had to consider carefully how to come up with reusable components to avoid creating frames repetitively.

3. Week 3: JavaScript and PHP Basics

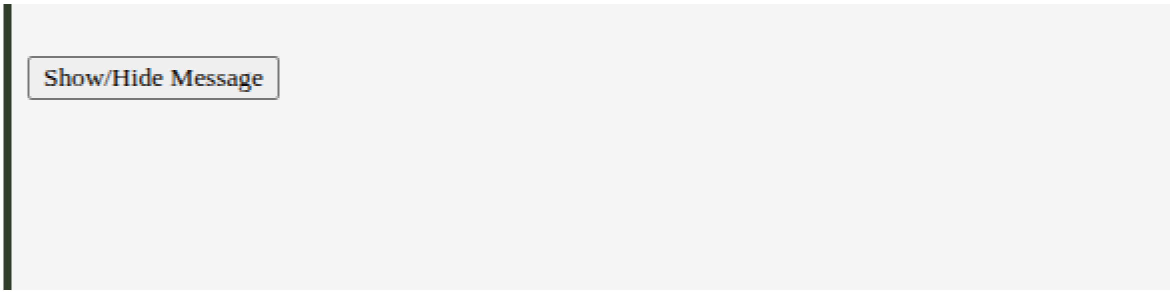
3.1. Figure 1: Invalid Login Validation



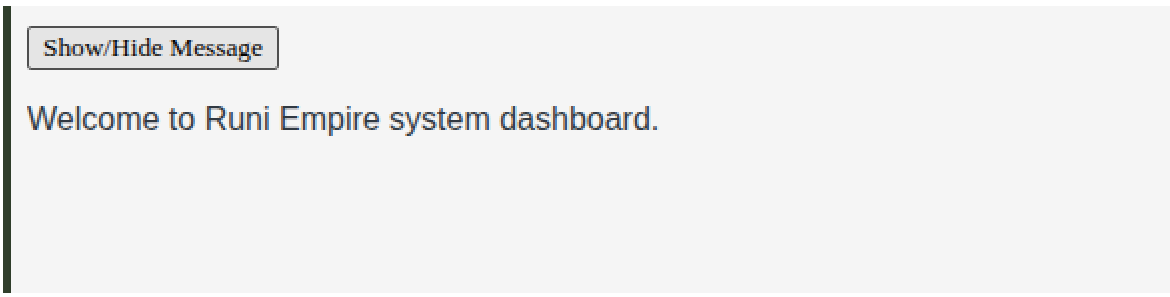
3.2. Fig 2: Valid login input accepted



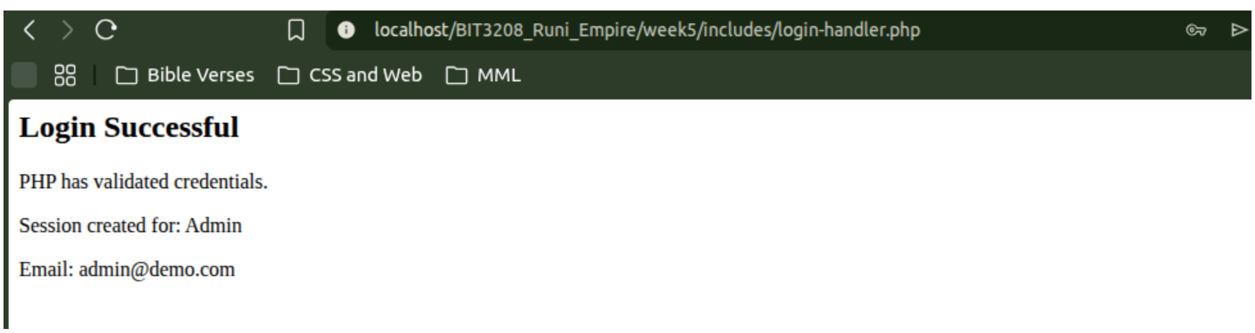
3.3. Fig 3: DOM manipulation interaction



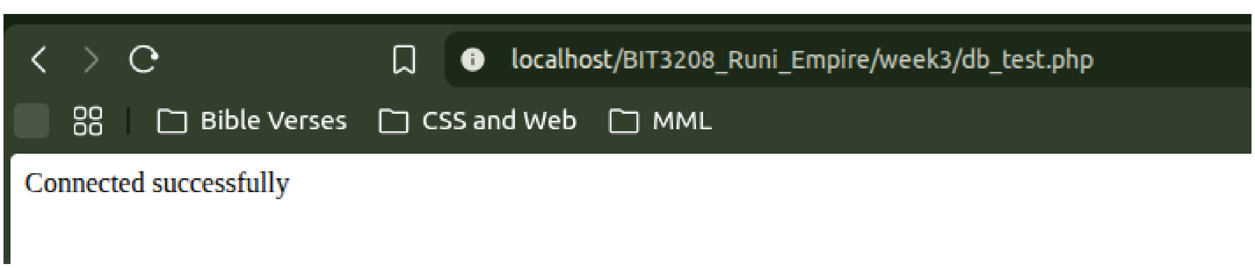
After button click



3.4. Figure 4: PHP Interaction Evidence after valid login details captured



3.5. Figure 5: Week 3 database connection success evidence

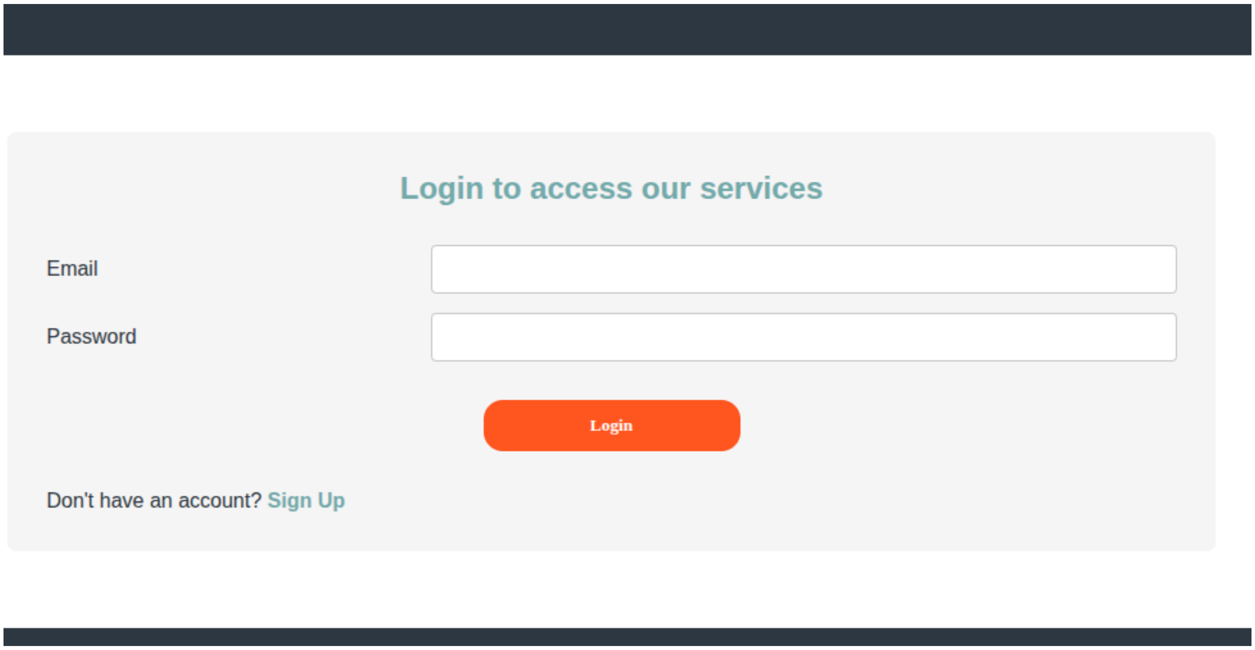


Week 3 Reflection:

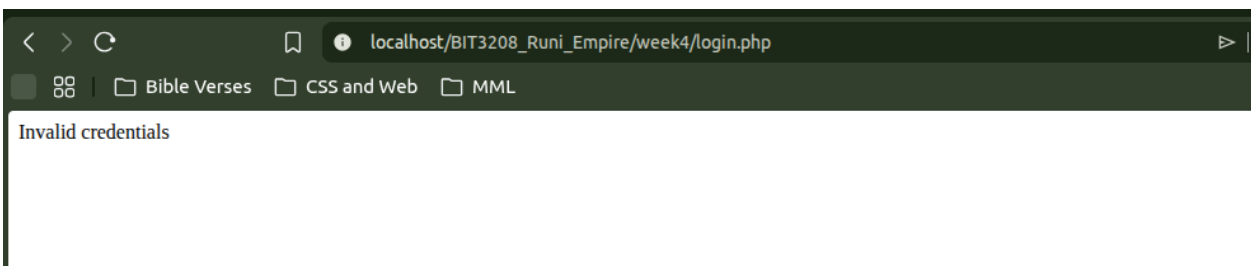
In week 3, much of the focus shifted to JavaScript and PHP processing. The most challenging aspect was using JavaScript to perform form validation, and to succeed, I had to learn a lot of concepts, mostly dealing with DOM Manipulation such as query selectors, creating nodes, and appending them to already existing parent containers.

4. WEEK 4: Server-Side Components and Backend Development Foundations

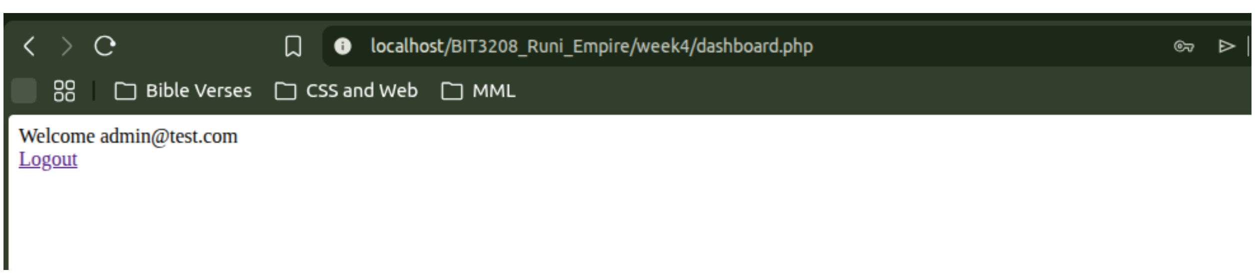
4.1. Figure 1: Login page initial state



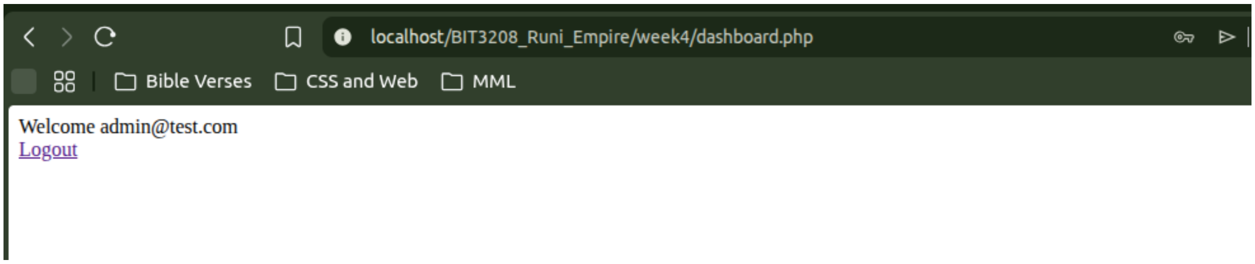
4.2. Figure 2: Failed Login Attempt



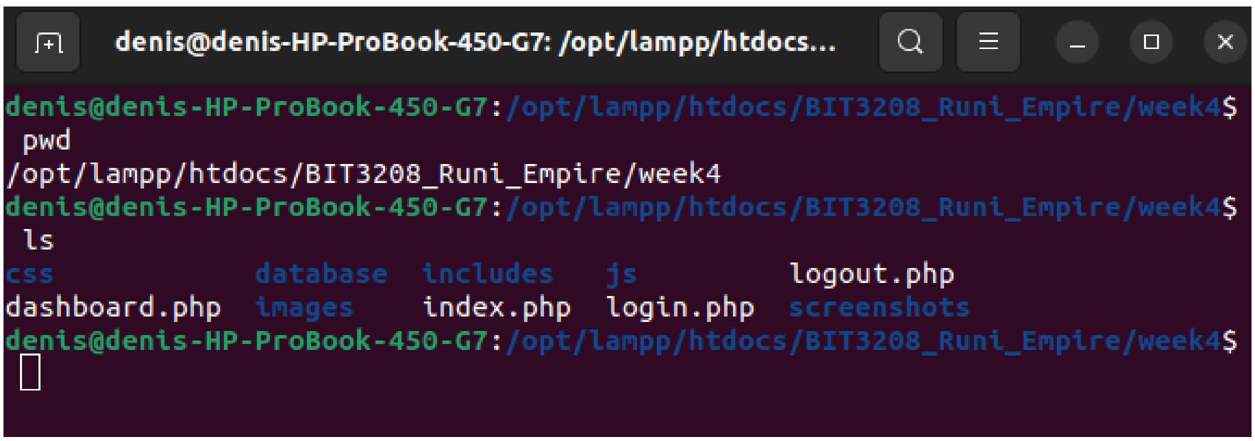
4.3. Figure 3: Successful login attempt - Redirected to dashboard



4.4. Figure 4: Session-based admin dashboard



4.5. Figure 5: Folder Structure



Week 4 Reflection:

In Week 4, the focus was on form processing, authentication basics, and sessions in PHP. The most difficult part was understanding how data moves from HTML forms into PHP and how validation is applied before processing. I learned how to capture form input using POST, validate it, and manage user sessions to maintain login state. I also built a basic login system and session-protected dashboard. This helped me understand how authentication systems control access to different parts of a web application.

5. Week 5: Database Components and CRUD Operations

5.1. Figure 5.1: Databases Screenshot

```
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| kinyozi_db         |
| mysql              |
| performance_schema |
| runi_empire_db     |
| school             |
| sys                |
| week1db            |
| week_one           |
+--------------------+
9 rows in set (0.01 sec)
```

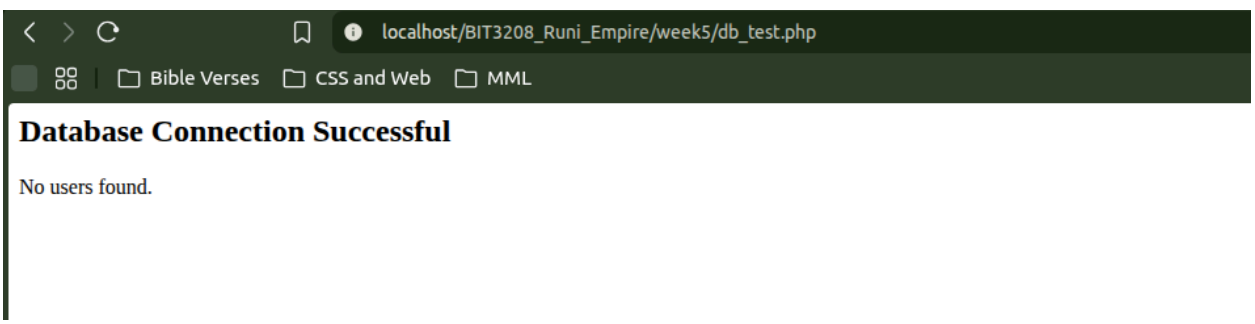
5.2. Figure 5.2: Created Tables within Database and defined their schema

```
mysql> USE runi_empire_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SHOW TABLES;
+--------------------------+
| Tables_in_runi_empire_db |
+--------------------------+
| bookings                 |
| users                    |
+--------------------------+
2 rows in set (0.00 sec)

mysql> DESCRIBE users;
+--------------+--------------+------+-----+-------------------+-------------------+
| Field        | Type         | Null | Key | Default           | Extra             |
+--------------+--------------+------+-----+-------------------+-------------------+
| id           | int          | NO   | PRI | NULL              | auto_increment    |
| first_name   | varchar(100) | YES  |     | NULL              |                   |
| last_name    | varchar(100) | YES  |     | NULL              |                   |
| email        | varchar(100) | YES  | UNI | NULL              |                   |
| phone_number | varchar(20)  | YES  |     | NULL              |                   |
| password     | varchar(255) | YES  |     | NULL              |                   |
| created_at   | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+--------------+--------------+------+-----+-------------------+-------------------+
7 rows in set (0.00 sec)
```

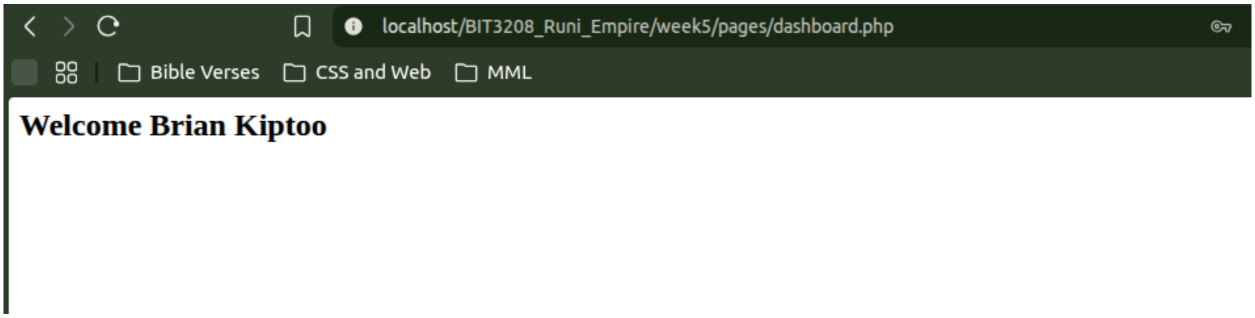
5.3. Figure 5.3: CRUD Operations on database

```
mysql> INSERT INTO users (first_name, last_name, email, phone_number, password)
    -> VALUES ('Brian', 'Kiptoo', 'brian.kiptoo@gmail.com', '0712345678', '1234');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM users;
+----+------------+-----------+------------------------+--------------+----------+---------------------+
| id | first_name | last_name | email                  | phone_number | password | created_at          |
+----+------------+-----------+------------------------+--------------+----------+---------------------+
|  1 | Brian      | Kiptoo    | brian.kiptoo@gmail.com | 0712345678   | 1234     | 2026-06-05 11:47:07 |
+----+------------+-----------+------------------------+--------------+----------+---------------------+
1 row in set (0.00 sec)

mysql> UPDATE users
    -> SET first_name = 'Kevin',
    ->     last_name = 'Mutua'
    -> WHERE id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM users;
+----+------------+-----------+------------------------+--------------+----------+---------------------+
| id | first_name | last_name | email                  | phone_number | password | created_at          |
+----+------------+-----------+------------------------+--------------+----------+---------------------+
|  1 | Kevin      | Mutua     | brian.kiptoo@gmail.com | 0712345678   | 1234     | 2026-06-05 11:47:07 |
+----+------------+-----------+------------------------+--------------+----------+---------------------+
1 row in set (0.00 sec)

mysql> DELETE FROM users
    -> WHERE id = 1;
Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM users;
Empty set (0.00 sec)

mysql> 
```

5.4. Figure 5.4: Database Connection



5.5. Figure 5.5: Successful login through the database



Week 5 Reflection:

In Week 5, I focused on databases and integrating them with PHP applications. The main challenge was understanding SQL operations and how PHP interacts with MySQL databases. I learned how to create databases and tables, perform CRUD operations, and connect PHP to a database using PDO. I also implemented database-driven login functionality and data retrieval. This improved my understanding of backend development and data management in web applications.